MULTIPLICITY OF A COMPONENT OF THE TOPOLOGICAL SPACE OF A STACK IS INDEPENDENT OF THE SMOOTH COVER

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original lemma 0DR5: sources/stacks-chow/tags.txt, source lines 17210–17210. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04XH: sources/stacks-chow/tags.txt, source lines 6395–6395. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DR6: sources/stacks-chow/tags.txt, source lines 17211–17211. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $U_1 \to \mathcal{X}$ and $U_2 \to \mathcal{X}$ be two smooth morphisms from schemes to a locally Noetherian algebraic stack $\mathcal{X}$. Let T_1' and T_2' be irreducible components of $|U_1|$ and $|U_2|$ respectively. Assume the closures of the images of T_1' and T_2' are the same irreducible component T of $|\mathcal{X}|$. Then $m_{T_1',U_1} = m_{T_2',U_2}$.*

Proof. Let V_1 and V_2 be dense subsets of T_1' and T_2', respectively, that are open in U_1 and U_2 respectively (see proof of Lemma 0DR5). The images of $|V_1|$ and $|V_2|$ in $|\mathcal{X}|$ are non-empty open subsets of the irreducible subset T, and therefore have non-empty intersection. By Properties of Stacks, Lemma 04XH, the map $|V_1 \times_{\mathcal{X}} V_2| \to |V_1| \times_{|\mathcal{X}|} |V_2|$ is surjective. Consequently $V_1 \times_{\mathcal{X}} V_2$ is a non-empty algebraic space; we may therefore choose an étale surjection $V \to V_1 \times_{\mathcal{X}} V_2$ whose source is a (non-empty) scheme. If we let T' be any irreducible component of V, then Lemma 0DR5 shows that the closure of the image of T' in U_1 (respectively U_2) is equal to T_1' (respectively T_2').

Applying Lemma 0DR6 twice we find that

$$m_{T_1',U_1} = m_{T',V} = m_{T_2',U_2},$$

as required. □

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